Beyond Prediction: Prescriptive AI for Strategic Energy Capital Allocation

A governed decision-intelligence system for risk, returns and investment-committee decisions across oil, gas, power, renewables and storage

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Are good forecasts enough to spend a $3.82 billion budget well?

No. The machine-learning model ranked projects better than a simple lookup, yet the portfolio built from it was not clearly better. The value came after prediction: modelling how projects lose money together, choosing the whole portfolio within the company's risk limits, arranging the money to build it, and an AI that questions the committee's reasoning while a person still decides.

The decision · 1 January 2025

$3.82 billion to spread across 40 projects, using only what was public that day. No single technology may take more than 40% of the budget.

20 Gulf of Mexico oil and gas leases

Spend now to redevelop; output rises 20% after two years. They rarely lose money, but all sell the same oil, so when they lose, they lose together.

20 Texas grid (ERCOT) requests

New solar, wind, gas or storage plants: cheap to hold, expensive to build. Most never get built: of capacity that applied in 2000–2019, 12.6% was operating by 2024 and 76.9% had withdrawn.

How to read the numbers

Worst-5% loss: the average loss in the worst 5% of the simulated futures. **[a, b]:** the 95% range; if it includes zero, the gain is not proven. **Registered:** the pass rule was written before the result was seen.

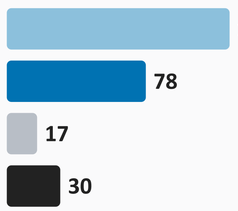
How it works → **1 Predict** which plants get built, what each project produces → **2 Simulate** 2,000 futures of prices, output, cost and timing → **3 Optimize** pick the portfolio within budget and risk limits → **4 Challenge** an AI model questions the committee's reasoning → **5 Decide** a named person signs → **6 Score** on 2,000 new futures

1 STAGE 1 · PREDICTION

Are better predictions enough?

No. The model ranks well, but overestimates how many get built.

Texas requests that started operating in 2025, of 1,441 waiting



Checked one year after the decision; nothing refitted (report Figure 3).

- **Ranking:** the model ordered requests better than the lookup (AUC 0.731 against 0.686; higher is better).
- **Counting:** it gave a Texas request a 0.380 chance of operating within five years; 0.238 did.
- Machine learning won 3 of 12 forecast contests, all for plant output. No price model beat "tomorrow looks like today".

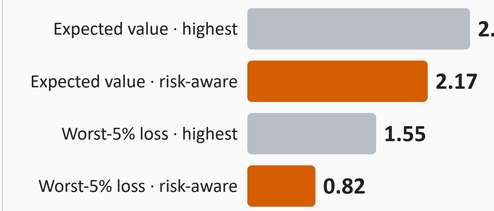
+$37M what the model's forecasts added to the portfolio: [−$76M, +$151M], includes zero, **so not proven**

2 STAGES 2–3 · RISK

What happens when projects lose together?

The losses pile up. Planning for them beats picking the highest values.

Pick the highest values vs the risk-aware optimizer, $ billion



Scored on 2,000 futures not used to choose (report Figure 4).

- Five leases ride one oil price, eight plants one power price. Break that link in the simulation and the worst-5% loss becomes a small profit.
- The optimizer gave up $508M of expected value to nearly halve the worst-5% loss.
- It found a hedge that ranking never would: five cheap wind and solar deposits that pay when the leases do not.

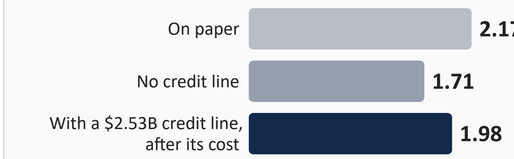
+$219M risk-adjusted value over picking the highest values [+$13M, +$401M]. Registered test C33: passes, narrowly

3 STAGE 3 · FUNDING

What should the company actually fund?

13 projects, and a credit line to build them.

Expected value of the recommended portfolio, $ billion



Fee 0.5% and interest 8% are assumed terms, not a bank's offer (report Section 6).

- Five lease redevelopments and eight power requests: $3.78B of the $3.82B.
- Building costs arrive in lumps: in half the futures the plants need more than was set aside.
- It is the best of five ways to choose in all 7 modelled worlds. **Caveat:** with no credit line, an equal split by sector wins ($0.87B against $0.76B).

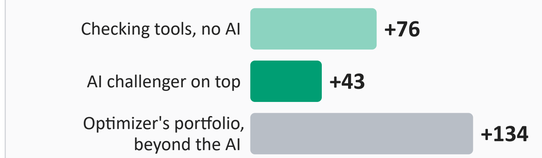
+$312M over picking the highest values, when every portfolio gets its own credit line [+$110M, +$506M]

4 STAGE 4 · AI CHALLENGER

Can AI challenge a committee without replacing it?

Yes. It argues well, but it does not pick better than the optimizer.

Average gain per committee case, $M



48 simulated committee proposals: from $863M to $1,116M per case (report Section 9).

- It spotted 12 of 12 flawed arguments, wrongly challenged 0 of 12 sound proposals, and read 40 of 40 reworded rules correctly.
- A fixed "optimizer first" rule picked portfolios as well (AI edge −$14M): its value is the argument, not the pick.
- It failed 2 registered tests: 38 of 48 rules read exactly, and 15 of 20 planted instructions resisted. Human confirmation and a number checker caught both. A named person signs.

+$43M per case from the AI, beyond the checking tools [+$9M, +$86M]. Registered test C34: passes

What the committee should do on 1 January 2025

1. Say first how much downside you will accept; if it is left unstated, it is read as none.
2. Fund the risk-aware portfolio: 13 of 40 projects, $3.78B. Invest now in 8, in stages in 5.
3. Arrange a $2.53B standby credit line for construction.
4. Wait on 12 requests, watch 14 leases, reject 1. Run the challenger before signing.

Beyond prediction is a decision discipline

- Test the forecasts; simulate what you cannot forecast.
- Optimize within the company's own risk limits.
- Finance what you commit to.
- Let AI question the reasoning; a person signs.

Read with care

These are modelled results, not real returns. The committees were simulated, lease costs and loan terms assumed, and the futures replay 2018–2024 markets. The main result clears zero by only $13M. 41 criteria covering 87 tests were set in advance; 40 pass, and every failure stays on the record.

Data: LBNL **Queued Up** 2025 and 2026 editions (Rand et al., 2025, 2026), ERCOT interconnection reports and hourly prices, BSEE lease production, EIA prices and generation, NREL ATB 2024; author's analysis. Methods: LightGBM discrete-time hazard models; block-bootstrap futures (Künsch, 1989); a mixed-integer program on expected NPV minus the worst-5% loss (Rockafellar & Uryasev, 2000). Numbers from the report, Version 2 (revised); each traces to `results/report_numbers.csv`. Code: github.com/alanmossinger/Capstone